BUSINESS INTELLIGENCE

DATA MODELLING

KHEN MUHAMMAD CAHYO

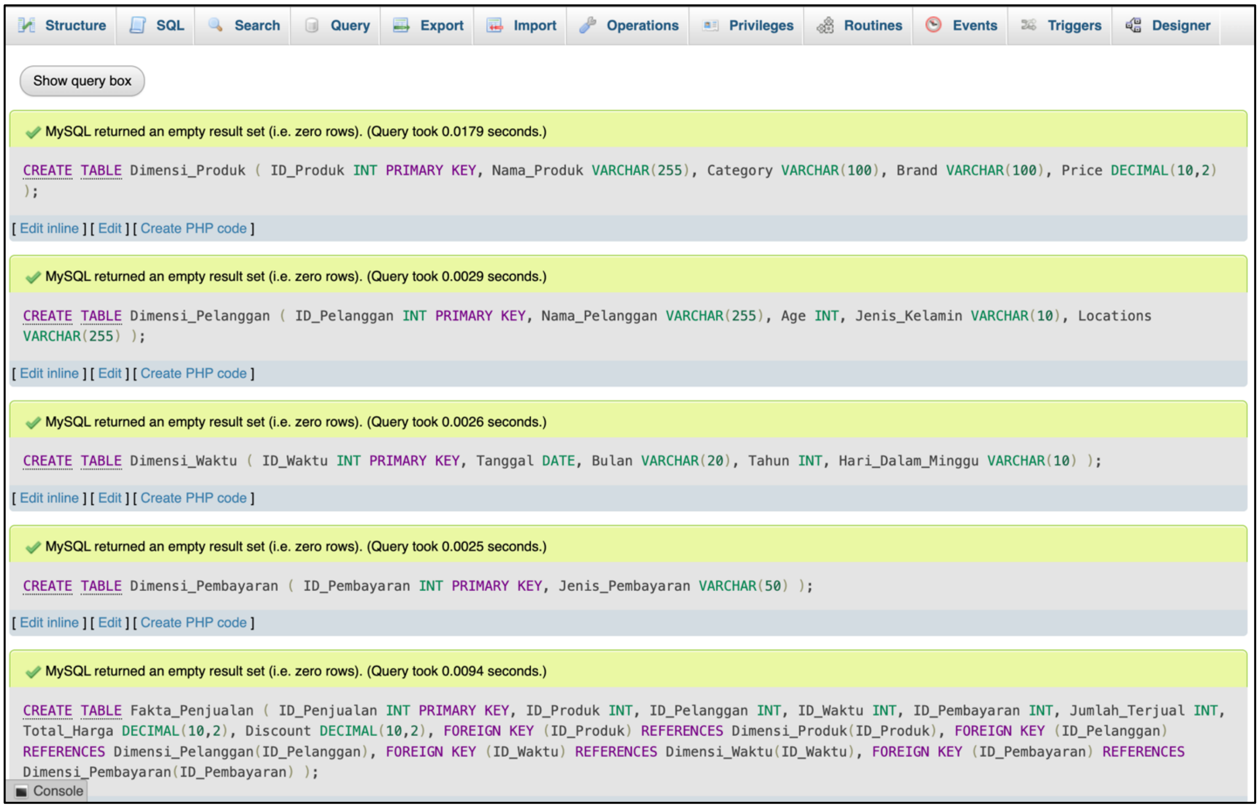
244107023002

TI-3I

STATE POLYTECHNIC OF MALANG

INFORMATION TECHNOLOGY DEPARTMENT

TASK NUMBER 1 – CREATE DATABASE toko_maju



TASK NUMBER 2 – ENTER A DATA ON EACH TABLE

- **Insert Into Dimensi Pembayaran**

Show query box

✔ 10 rows inserted. (Query took 0.0022 seconds.)

```
INSERT INTO Dimensi_Pembayaran (ID_Pembayaran, Jenis_Pembayaran) VALUES (1, 'Kartu Kredit'), (2, 'Transfer Bank'), (3, 'e-Wallet'), (4, 'COD'),
(5, 'Virtual Account'), (6, 'Debit Card'), (7, 'PayLater'), (8, 'Kartu Debit'), (9, 'QRIS'), (10, 'Paypal');
```

[Edit inline] [Edit] [Create PHP code]

- **Insert Into Dimensi Pelanggan**

Show query box

✔ 10 rows inserted. (Query took 0.0006 seconds.)

```
INSERT INTO Dimensi_Pelanggan (ID_Pelanggan, Nama_Pelanggan, Age, Jenis_Kelamin, Locations) VALUES (1, 'Ahmad', 30, 'Laki-laki', 'Jakarta'), (2,
'Siti', 25, 'Perempuan', 'Bandung'), (3, 'Budi', 40, 'Laki-laki', 'Surabaya'), (4, 'Rina', 35, 'Perempuan', 'Yogyakarta'), (5, 'Tono', 28, 'Laki-
laki', 'Semarang'), (6, 'Dewi', 27, 'Perempuan', 'Jakarta'), (7, 'Rahmat', 34, 'Laki-laki', 'Surabaya'), (8, 'Indah', 29, 'Perempuan', 'Bandung'),
(9, 'Hendra', 42, 'Laki-laki', 'Medan'), (10, 'Sari', 31, 'Perempuan', 'Yogyakarta');
```

[Edit inline] [Edit] [Create PHP code]

- **Insert Into Dimensi Produk**

Show query box

✔ 10 rows inserted. (Query took 0.0007 seconds.)

```
INSERT INTO Dimensi_Produk (ID_Produk, Nama_Produk, Category, Brand, Price) VALUES (1, 'Laptop A', 'Elektronik', 'BrandX', 10000000.00), (2,
'Smartphone B', 'Elektronik', 'BrandY', 5000000.00), (3, 'Kipas Angin C', 'Peralatan Rumah Tangga', 'BrandZ', 750000.00), (4, 'TV LED D',
'Elektronik', 'BrandW', 4000000.00), (5, 'Mesin Cuci E', 'Peralatan Rumah Tangga', 'BrandY', 3500000.00), (6, 'Headphone F', 'Elektronik',
'BrandZ', 1500000.00), (7, 'Kompor Gas G', 'Peralatan Rumah Tangga', 'BrandZ', 1200000.00), (8, 'Smartwatch H', 'Elektronik', 'BrandX',
2000000.00), (9, 'Blender I', 'Peralatan Rumah Tangga', 'BrandW', 850000.00), (10, 'Kamera DSLR J', 'Elektronik', 'BrandV', 7500000.00);
```

[Edit inline] [Edit] [Create PHP code]

- **Insert Into Dimensi Waktu**

Show query box

✔ 10 rows inserted. (Query took 0.0008 seconds.)

```
INSERT INTO Dimensi_Waktu (ID_Waktu, Tanggal, Bulan, Tahun, Hari_Dalam_Minggu) VALUES (1, '2024-01-15', 'Januari', 2024, 'Senin'), (2, '2024-02-
10', 'Februari', 2024, 'Sabtu'), (3, '2024-03-05', 'Maret', 2024, 'Selasa'), (4, '2024-04-20', 'April', 2024, 'Jumat'), (5, '2024-05-12', 'Mei',
2024, 'Minggu'), (6, '2024-06-15', 'Juni', 2024, 'Sabtu'), (7, '2024-07-20', 'Juli', 2024, 'Minggu'), (8, '2024-08-08', 'Agustus', 2024, 'Kamis'),
(9, '2024-09-18', 'September', 2024, 'Rabu'), (10, '2024-10-25', 'Oktober', 2024, 'Jumat');
```

[Edit inline] [Edit] [Create PHP code]

- **Insert Into Fakta Penjualan**

Show query box

✔ 15 rows inserted. (Query took 0.0022 seconds.)

```
INSERT INTO Fakta_Penjualan (ID_Penjualan, ID_Produk, ID_Pelanggan, ID_Waktu, ID_Pembayaran, Jumlah_Terjual, Total_Harga, Discount) VALUES (1, 1,
1, 1, 2, 2, 20000000, 500000), (2, 2, 2, 2, 3, 1, 5000000, 250000), (3, 3, 3, 3, 4, 3, 2250000, 150000), (4, 4, 4, 4, 1, 1, 4000000, 200000), (5,
5, 5, 5, 5, 2, 7000000, 350000), (6, 6, 6, 6, 6, 3, 4500000, 300000), (7, 7, 7, 7, 7, 2, 2400000, 120000), (8, 8, 8, 8, 8, 1, 2000000, 150000),
(9, 9, 9, 9, 9, 4, 3400000, 200000), (10, 10, 10, 10, 10, 1, 7500000, 500000), (11, 1, 2, 3, 1, 2, 20000000, 750000), (12, 2, 4, 5, 2, 1, 5000000,
200000), (13, 3, 6, 7, 3, 5, 3750000, 300000), (14, 4, 8, 9, 4, 1, 4000000, 250000), (15, 5, 10, 1, 5, 3, 10500000, 500000);
```

[Edit inline] [Edit] [Create PHP code]

TASK NUMBER 3 – DATA ANALYSIS QUERIES EXAMPLE

1. Total Sales per Product Category

Showing rows 0 - 1 (2 total, Query took 0.0008 seconds.)

```
SELECT dp. Category, SUM(fp. Total_Harga) AS Total_Pendapatan FROM Fakta_Penjualan fp JOIN Dimensi_Produk dp ON fp.ID_Produk = dp.ID_Produk GROUP BY dp. Category;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Category	Total_Pendapatan
Elektronik	72000000.00
Peralatan Rumah Tangga	29300000.00

2. Number of Transactions per Payment Method

Showing rows 0 - 9 (10 total, Query took 0.0003 seconds.)

```
SELECT dp. Jenis_Pembayaran, COUNT(fp.ID_Penjualan) AS Jumlah_Transaksi FROM Fakta_Penjualan fp JOIN Dimensi_Pembayaran dp ON fp.ID_Pembayaran = dp.ID_Pembayaran GROUP BY dp. Jenis_Pembayaran;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Jenis_Pembayaran	Jumlah_Transaksi
Kartu Kredit	2
Transfer Bank	2
e-Wallet	2
COD	2
Virtual Account	2
Debit Card	1
PayLater	1
Kartu Debit	1
QRIS	1
Paypal	1

3. Monthly Sales Trends in 2024

Showing rows 0 - 9 (10 total, Query took 0.0005 seconds.)

```
SELECT dw.Bulan, dw.Tahun, SUM(fp.Total_Harga - fp.Discount) AS Total_Penjualan FROM Fakta_Penjualan fp JOIN Dimensi_Waktu dw ON fp.ID_Waktu = dw.ID_Waktu WHERE dw.Tahun = 2024 GROUP BY dw.Tahun, dw.Bulan ORDER BY MIN(dw.Tanggal);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Bulan	Tahun	Total_Penjualan
Januari	2024	29500000.00
Februari	2024	4750000.00
Maret	2024	21350000.00
April	2024	3800000.00
Mei	2024	11450000.00
Juni	2024	4200000.00
Juli	2024	5730000.00
Agustus	2024	1850000.00
September	2024	6950000.00
Oktober	2024	7000000.00

4. Total Sales per Month

Showing rows 0 - 9 (10 total, Query took 0.0003 seconds.)

```sql
SELECT dw.Bulan, dw.Tahun, SUM(fp.Total_Harga - fp.Discount) AS Total_Penjualan FROM Fakta_Penjualan fp JOIN Dimensi_Waktu dw ON fp.ID_Waktu = dw.ID_Waktu WHERE dw.Tahun = 2024 GROUP BY dw.Bulan, dw.Tahun ORDER BY MIN(dw.Tanggal);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Bulan	Tahun	Total_Penjualan
Januari	2024	29500000.00
Februari	2024	4750000.00
Maret	2024	21350000.00
April	2024	3800000.00
Mei	2024	11450000.00
Juni	2024	4200000.00
Juli	2024	5730000.00
Agustus	2024	1850000.00
September	2024	6950000.00
Oktober	2024	7000000.00

5. Product Categories with the Most Sales

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

```sql
SELECT dp.Category, SUM(fp.Jumlah_Terjual) AS Total_Jumlah, SUM(fp.Total_Harga - fp.Discount) AS Total_Penjualan FROM Fakta_Penjualan fp JOIN Dimensi_Produk dp ON fp.ID_Produk = dp.ID_Produk GROUP BY dp.Category ORDER BY Total_Penjualan DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Category	Total_Jumlah	Total_Penjualan 1
Elektronik	13	68900000.00
Peralatan Rumah Tangga	19	27680000.00

6. Most Frequently Used Payment Methods

Showing rows 0 - 9 (10 total, Query took 0.0003 seconds.)

```sql
SELECT dpay.Jenis_Pembayaran, COUNT(fp.ID_Penjualan) AS Jumlah_Transaksi, SUM(fp.Total_Harga - fp.Discount) AS Total_Penjualan FROM Fakta_Penjualan fp JOIN Dimensi_Pembayaran dpay ON fp.ID_Pembayaran = dpay.ID_Pembayaran GROUP BY dpay.Jenis_Pembayaran ORDER BY Jumlah_Transaksi DESC;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

Jenis_Pembayaran	Jumlah_Transaksi 1	Total_Penjualan
Kartu Kredit	2	23050000.00
Transfer Bank	2	24300000.00
e-Wallet	2	8200000.00
COD	2	5850000.00
Virtual Account	2	16650000.00
Debit Card	1	4200000.00
PayLater	1	2280000.00
Kartu Debit	1	1850000.00
QRIS	1	3200000.00
Paypal	1	7000000.00

7. **Customers with the Highest Total Purchases**

Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)

```
SELECT dpel.Nama_Pelanggan, SUM(fp.Total_Harga - fp.Discount) AS Total_Pembelian, COUNT(fp.ID_Penjualan) AS Jumlah_Transaksi FROM Fakta_Penjualan
fp JOIN Dimensi_Pelanggan dpel ON fp.ID_Pelanggan = dpel.ID_Pelanggan GROUP BY dpel.Nama_Pelanggan ORDER BY Total_Pembelian DESC LIMIT 5;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

Nama_Pelanggan	Total_Pembelian 1	Jumlah_Transaksi
Siti	24000000.00	2
Ahmad	19500000.00	1
Sari	17000000.00	2
Rina	8600000.00	2
Dewi	7650000.00	2

TASK NUMBER 4 – CONCLUSION

Based on the results of the lab work, we successfully built a simple data warehouse with dimension tables (product, customer, time, payment) and a sales fact table. The analysis query reveals that monthly sales trends in 2024 were quite varied, with a peak in January (Rp 29,500,000) and a decline in August (Rp 1,850,000). Product category analysis shows that the Electronics category generated the largest total sales, while the Household Appliances category outperformed in terms of units sold. In terms of payment methods, no single method was dominant, but Bank Transfer, Credit Card, e-Wallet, Cash on Delivery (COD), and Virtual Account emerged as the most frequently used methods.

Furthermore, customer analysis shows that several customers are key contributors to total purchases, such as Siti, Ahmad, and Sari, who account for large transactions. This underscores the importance of applying multidimensional analysis to understand consumer behavior, sales patterns, and the effectiveness of payment methods. With this approach, companies can more easily make strategic decisions, such as determining marketing strategies, providing special offers to potential customers, and optimizing the most popular payment methods.